

Multimodal Large Model+Color Blocks Back In Place

Function Description

After the program runs, after waking up the voice module, input the command to remember the current positions of the color blocks by voice. After the program finishes recording, shuffle the positions of the color blocks. Then wake up the voice module again and input by voice to end the current recording task. Then wake up the voice module again and input the color blocks return-to-place command by voice. After the large model thinks it through, it will plan an action list, and the program will control the robotic arm to complete the color blocks return-to-place task based on the action functions in the action list.

Startup

Open the terminal and enter the following command:

```
ros2 launch largemodel largemodel_control.launch.py
```

After waking up the voice module, input by voice:

```
Remember the current positions of the color blocks
```

After recording the color block positions, after waking up the voice module, input by voice:

```
End the current recording task
```

Then you can shuffle the positions of the color blocks. After shuffling, after waking up the voice module, input by voice:

```
Return the blocks on the desktop to their original positions
```

After the buzzer beeps once, it starts gripping the color blocks according to the order in which the color block positions were recorded in the program and placing them back to their original positions. If there are other color blocks on top of the color block that needs to be returned to its original position, it will first remove the color blocks on top, then grip and place back to the original position.

Task Planning

Record Current Positions of Color Blocks

1. Call `seewhat()` to observe the environment;
2. Call `seewhat()` again to observe the environment, then determine which color block is the top block;
3. If there are two or more piles of color blocks in the environment, select and execute the following content based on whether a color block is a single one. Calculate for each pile of color blocks. Only one action function can be output at a time here:
 - (1) If it is a single color block, pass the outer frame coordinates of the color block into the function `compute_pose(x1, y1, x2, y2, 'name')`, where `name` represents the color of the color block, and its value can be `red`, `blue`, `green`, or `yellow`;

- (2) If it is not a single color block, pass the outer frame coordinates of the top color block into the function `compute_pose_order(x1, y1, x2, y2, 'name', 'order')`, where `name` represents the color of the color block (with values `red`, `blue`, `green`, or `yellow`), and `order` represents the top-to-bottom arrangement order of the color blocks. Its value is a string composed of the initial letters of the colors of each layer of color blocks from top to bottom. For example, if the top-to-bottom arrangement order of the color blocks is red, blue, green, yellow, then the value of `order` is `rbgy`, and so on.
4. If there is only one pile of color blocks, pass the outer frame coordinates of the top color block into the function `compute_pose_order(x1, y1, x2, y2, 'name', 'order')`, where `name` represents the color of the color block (with values `red`, `blue`, `green`, or `yellow`), and `order` represents the top-to-bottom arrangement order of the color blocks. Its value is a string composed of the initial letters of the colors of each layer of color blocks from top to bottom. For example, if the top-to-bottom arrangement order of the color blocks is red, blue, green, yellow, then the value of `order` is `rbgy`, and so on.

Color Blocks Back In Place

1. First, call the `color_back_to_orin()` function to control the color blocks to return to their original positions.
2. Call the `check_remove(color)` function to check whether the target color block exists in the removal list (where `color` is the color of the target color block, and its value range is `red`, `blue`, `green`, or `yellow`).
3. If the target color block exists in the removal list, perform the following operations:
 - (1) Call the `grasp_from_rm_list(color)` function to grasp the target color block from the removal list (the meaning of the `color` parameter is the same as above);
 - (2) Pass the color of the target color block as a parameter into the `return_to_orin(color)` function. For example, when the target color block is red, call `return_to_orin(red)` (the value range of the `color` parameter is the same as above).
4. If the target color block does not exist in the removal list, follow the steps below to put the color block back to its original position (only one action function can be output at a time):
 - (1) Call the `seewhat()` function to observe the environment;
 - (2) Determine whether there are other color blocks on top of the target color block that needs to be returned to its original position;
 - (3) If there are no other color blocks on top of the target color block that needs to be returned to its original position, perform the following steps:
 - ① Call the `grasp_obj(x1, y1, x2, y2)` function to grasp the target color block that needs to be returned to its original position (where `(x1, y1, x2, y2)` are the border coordinates of the top surface of the target color block that needs to be returned to its original position);
 - ② Pass the color of the target color block as a parameter into the `return_to_orin(color)` function (the value range of the `color` parameter is the same as above).
 - (4) If there are other color blocks on top of the target color block that needs to be returned to its original position, perform the following steps:
 - ① Call the `remove_obj(x1, y1, x2, y2, color)` function to remove the color block (where `(x1, y1, x2, y2)` are the border coordinates of the top surface of this color block, and `color` is the color of this color block, with a value range of `red`, `blue`, `green`, or `yellow`);
 - ② Call the `seewhat()` function to observe the environment;
 - ③ If there are no other color blocks above the target color block that needs to be returned to its original position, call the `grasp_obj(x1, y1, x2, y2)` function to grasp the target color block (where

(x1, y1, x2, y2) are the border coordinates of the top surface of the target color block);

④Pass the color of the target color block as a parameter into the `return_to_orin(color)` function (the value range of the `color` parameter is the same as above).

Core Code Analysis

Please refer to the **Core Code Analysis** content in the tutorial **【AI Model-Text Version】 - 【Multimodal Large Model + Color Blocks Back In Place】** . The voice version and text version have the same action functions; only the input method for task commands is different.